

Assignment 4 Solutions

$$1. \text{ fun } f \rightarrow \text{ fun } a \rightarrow f (\text{ fun } g \rightarrow g a)$$

2.

$$\frac{\frac{\frac{}{\text{foo} \Diamond \text{bar}} \text{ (fooBar)}}{\text{foo} \Diamond \text{bar}} \quad \frac{\frac{\frac{}{\text{foo} \Diamond \text{bar}} \text{ (fooBar)}}{\text{foo} \Diamond \text{bar}} \quad \frac{\frac{\frac{}{\text{cat} \Diamond \text{dog}} \text{ (catDog)}}{\text{dog} \Box \text{cat}} \text{ (swap)}}{\text{foo cat} \Box \text{bar dog}} \text{ (combine)}}{\text{foo bar dog} \Box \text{bar foo cat}} \text{ (combine)}$$

3.

$$\frac{\frac{\frac{\frac{}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash g: \text{int} \rightarrow \text{bool}} \text{ (var)}}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash g \text{ } 0: \text{int}} \text{ (init)}}{\frac{\frac{\frac{}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash f: \text{bool} \rightarrow \text{int}} \text{ (var)}}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash f (g \text{ } 0): \text{int}} \text{ (app)}}{\frac{\frac{\frac{}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash g: \text{int} \rightarrow \text{bool}} \text{ (var)}}{\{f: \text{bool} \rightarrow \text{int}, g: \text{int} \rightarrow \text{bool}\} \vdash g (f (g \text{ } 0)): \text{bool}} \text{ (app)}} \text{ (app)}$$

bool can be replaced with any other type in the above derivation.

4.

$$\frac{\frac{\frac{\frac{}{\{x: \text{bool}\} \vdash \text{false}: \text{bool}} \text{ (false)}}{\{x: \text{bool}\} \vdash \text{false} \parallel x: \text{bool}} \text{ (var)}}{\frac{\frac{\frac{}{\{x: \text{bool}\} \vdash 3: \text{int}} \text{ (true)}}{\{x: \text{bool}\} \vdash 3 + (\text{if false} \parallel x \text{ then } 2 \text{ else } 1): \text{int}} \text{ (init)}}{\frac{\frac{}{\{ \} \vdash \text{true}: \text{bool}} \quad \frac{\frac{\frac{}{\{x: \text{bool}\} \vdash 3: \text{int}} \text{ (true)}}{\{x: \text{bool}\} \vdash 3 + (\text{if false} \parallel x \text{ then } 2 \text{ else } 1): \text{int}} \text{ (init)}}{\{ \} \vdash \text{let } x = \text{true} \text{ in } 3 + (\text{if false} \parallel x \text{ then } 2 \text{ else } 1): \text{int}} \text{ (let)}} \text{ (let)}$$

		$\frac{\text{false} \Downarrow \textcolor{violet}{1} \quad \text{true} \Downarrow \textcolor{violet}{T}}{\text{false} \parallel \text{true} \Downarrow \textcolor{violet}{T}} \quad \textcolor{red}{(false)} \quad \textcolor{red}{(trueE)}$		
		$\frac{\textcolor{red}{(falseE)} \quad \text{false} \parallel \text{true} \Downarrow \textcolor{violet}{T} \quad \textcolor{red}{(orE)} \quad \frac{2 \Downarrow 2}{\text{if false} \parallel \text{true} \text{ then } 2 \text{ else } 1 \Downarrow \textcolor{violet}{2}} \quad \textcolor{red}{(intLitE)}$		
$\text{true} \Downarrow \textcolor{violet}{T} \quad \textcolor{red}{(trueE)}$	$3 \Downarrow \textcolor{violet}{3}$	$\frac{\text{if false} \parallel \text{true} \text{ then } 2 \text{ else } 1 \Downarrow \textcolor{violet}{2}}{3 + \text{if false} \parallel \text{true} \text{ then } 2 \text{ else } 1 \Downarrow \textcolor{violet}{5}} \quad \textcolor{red}{(ifE)} \quad \textcolor{red}{(addIntE)}$		
$\frac{\text{true} \Downarrow \textcolor{violet}{T} \quad \textcolor{red}{(trueE)} \quad \frac{3 \Downarrow \textcolor{violet}{3} \quad \text{if false} \parallel \text{true} \text{ then } 2 \text{ else } 1 \Downarrow \textcolor{violet}{2}}{3 + \text{if false} \parallel \text{true} \text{ then } 2 \text{ else } 1 \Downarrow \textcolor{violet}{5}} \quad \textcolor{red}{(letE)}$				
$\text{let } x = \text{true} \text{ in } 3 + (\text{if false} \parallel x \text{ then } 2 \text{ else } 1) \Downarrow \textcolor{violet}{5}$				

5.

$$\begin{array}{c}
 \frac{\frac{\frac{\{f: \text{int} \rightarrow \text{bool}\} \vdash f: \text{int} \rightarrow \text{bool} \quad \{f: \text{int} \rightarrow \text{bool}\} \vdash z: \text{int}}{\{f: \text{int} \rightarrow \text{bool}\} \vdash f \ z: \text{bool}} \quad \frac{\{f: \text{int} \rightarrow \text{bool}, x: \text{int}\} \vdash \text{true}: \text{bool}}{\{f: \text{int} \rightarrow \text{bool}\} \vdash \text{fun } x \rightarrow \text{true}: \text{int} \rightarrow \text{bool}}}{\{f: \text{int} \rightarrow \text{bool}\} \vdash \text{if } f \ z \text{ then fun } x \rightarrow \text{true} \text{ else } f: \text{int} \rightarrow \text{bool}} \quad \frac{\{f: \text{int} \rightarrow \text{bool}\} \vdash f: \text{int} \rightarrow \text{bool} \quad \{f: \text{int} \rightarrow \text{bool}\} \vdash z: \text{int}}{\{f: \text{int} \rightarrow \text{bool}\} \vdash (f \ z \text{ then } (\text{fun } x \rightarrow \text{true}) \text{ else } f) \ z: \text{bool}}
 \end{array}$$

(ver) (intLit) (true) (fun) (var) (if) (intLit) (app)